

36.1 Baseline-category logits

Let the response of observation i fall in one of c categories, recorded as a vector of counts $\mathbf{Y}_i = (Y_{i1}, \dots, Y_{ic})^\top$ with $\sum_r Y_{ir} = m_i$. When each row is a single subject, $m_i = 1$ and $\mathbf{Y}_i$ is an indicator vector; when rows are covariate patterns, m_i counts the subjects sharing that pattern. As with grouped and ungrouped binary data in [Chapter 35](#), the two cases are the same model with different bookkeeping, and the distinction matters only for the deviance as a goodness-of-fit statistic.

36.1.1 The multinomial distribution is an exponential family

[Chapter 34](#) was built on a one-dimensional exponential dispersion family (Definition 34.1.1). A categorical response is a *vector* with $c - 1$ free coordinates, and the structure survives the move, with the natural parameter and the cumulant function becoming vector-valued.

Lemma 36.1.1 — The multinomial as an exponential family

Let $\mathbf{Y} \sim \text{Mult}(m, \boldsymbol{\pi})$ with $\pi_r > 0$ for every r and $\sum_{r=1}^c \pi_r = 1$. Write $\tilde{\mathbf{Y}} = (Y_1, \dots, Y_{c-1})^\top$ for the first $c - 1$ counts and

$$\eta_r = \log \frac{\pi_r}{\pi_c}, \quad r = 1, \dots, c - 1.$$

Then the probability mass function is

$$f(\mathbf{y}) = h(\mathbf{y}) \exp\{\tilde{\mathbf{y}}^\top \boldsymbol{\eta} - m b(\boldsymbol{\eta})\}, \quad b(\boldsymbol{\eta}) = \log\left(1 + \sum_{s=1}^{c-1} e^{\eta_s}\right), \quad (36.1.1)$$

with $h(\mathbf{y}) = m!/(y_1! \cdots y_c!)$. The map $\boldsymbol{\eta} \mapsto \boldsymbol{\pi}$ is a bijection from $\mathbb{R}^{c-1}$ onto the interior of the probability simplex, with inverse

$$\pi_r = \frac{e^{\eta_r}}{1 + \sum_{s < c} e^{\eta_s}} \quad (r < c), \quad \pi_c = \frac{1}{1 + \sum_{s < c} e^{\eta_s}}, \quad (36.1.2)$$

and the first two moments are

$$\mathbb{E}(\tilde{\mathbf{Y}}) = m \nabla b(\boldsymbol{\eta}) = m \tilde{\boldsymbol{\pi}}, \quad \text{Cov}(\tilde{\mathbf{Y}}) = m \nabla^2 b(\boldsymbol{\eta}) = m(\text{diag}(\tilde{\boldsymbol{\pi}}) - \tilde{\boldsymbol{\pi}} \tilde{\boldsymbol{\pi}}^\top), \quad (36.1.3)$$

where $\tilde{\boldsymbol{\pi}} = (\pi_1, \dots, \pi_{c-1})^\top$.

Proof. Since $\sum_r y_r = m$,

$$\prod_{r=1}^c \pi_r^{y_r} = \pi_c^m \prod_{r < c} \left(\frac{\pi_r}{\pi_c} \right)^{y_r} = \exp \left\{ \sum_{r < c} y_r \eta_r + m \log \pi_c \right\},$$

and $\log \pi_c = -b(\boldsymbol{\eta})$ by the definition of η_r , which gives [Eq. 36.1.1](#). Solving $\eta_r = \log \pi_r - \log \pi_c$ with $\sum_r \pi_r = 1$ gives [Eq. 36.1.2](#), and every $\boldsymbol{\pi}$ with positive entries arises from exactly one $\boldsymbol{\eta}$, so the map is a bijection.

For the moments, sum [Eq. 36.1.1](#) over the support. By the multinomial theorem,

$$M(\boldsymbol{\eta}) = \sum_{\mathbf{y}} h(\mathbf{y}) \exp(\tilde{\mathbf{y}}^\top \boldsymbol{\eta}) = \left(1 + \sum_{s < c} e^{\eta_s} \right)^m = e^{mb(\boldsymbol{\eta})},$$

a finite sum, which may therefore be differentiated term by term without any regularity condition. Since $\log M = mb$ is the cumulant generating function of $\tilde{\mathbf{Y}}$ shifted to $\boldsymbol{\eta}$, its first two derivatives are $E(\tilde{\mathbf{Y}}) = m \nabla b$ and $\text{Cov}(\tilde{\mathbf{Y}}) = m \nabla^2 b$. Finally

$$\frac{\partial b}{\partial \eta_r} = \frac{e^{\eta_r}}{1 + \sum_{s < c} e^{\eta_s}} = \pi_r, \quad \frac{\partial^2 b}{\partial \eta_r \partial \eta_s} = \pi_r (\delta_{rs} - \pi_s),$$

where δ_{rs} is one if $r = s$ and zero otherwise. ■

Remark (What is new and what is not). [Eq. 36.1.1](#) has the shape of Definition 34.1.1 with the scalar natural parameter replaced by a vector and the dispersion ϕ/w_i by $1/m$: the number of trials plays the role of a prior weight. The identity “mean is the gradient of the cumulant function, variance its second derivative” is Proposition 34.1.2, proved there by the same argument. New is only that $\nabla^2 b$ is a matrix, and a singular one once the baseline coordinate is put back: $\text{diag}(\boldsymbol{\pi}) - \boldsymbol{\pi} \boldsymbol{\pi}^\top$ kills $\mathbf{1}$, because the c counts sum to the fixed total m .

36.1.2 The model

Definition 36.1.1 — Baseline-category logit model

Let $\mathbf{Y}_1, \dots, \mathbf{Y}_n$ be independent with $\mathbf{Y}_i \sim \text{Mult}(m_i, \boldsymbol{\pi}_i)$, and let $\mathbf{x}_{(i)} \in \mathbb{R}^p$ be the i th row of a model matrix $\mathbf{X}$ whose first column is $\mathbf{1}$. The **baseline-category logit model** with baseline c is

$$\log \frac{\pi_{ir}}{\pi_{ic}} = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}_r, \quad r = 1, \dots, c-1, \quad (36.1.4)$$

with unknown $\boldsymbol{\beta}_1, \dots, \boldsymbol{\beta}_{c-1} \in \mathbb{R}^p$. Equivalently, writing $\boldsymbol{\beta}_c = \mathbf{0}$, $\pi_{ir} = \exp(\mathbf{x}_{(i)}^\top \boldsymbol{\beta}_r) / \sum_{s=1}^c \exp(\mathbf{x}_{(i)}^\top \boldsymbol{\beta}_s)$ for every r .

This is a generalized linear model in the sense of Definition 34.2.1, with the multinomial random component of [Lemma 36.1.1](#) and the *canonical* link, since [Eq. 36.1.4](#) sets η_{ir} itself equal to a linear predictor. The one departure from [Chapter 34](#) is that each observation carries a vector of

natural parameters, so the model's coefficient vector is the stack $\boldsymbol{\beta} = (\boldsymbol{\beta}_1^\top, \dots, \boldsymbol{\beta}_{c-1}^\top)^\top$ of length $(c-1)p$. Every regressor buys $c-1$ coefficients: the price of treating the categories as unordered names.

36.1.3 The likelihood equations

Theorem 36.1.2 — Score, information and concavity

Under Definition 36.1.1 the log-likelihood, as a function of $\boldsymbol{\beta} = (\boldsymbol{\beta}_1^\top, \dots, \boldsymbol{\beta}_{c-1}^\top)^\top$, is

$$\ell(\boldsymbol{\beta}) = \sum_{i=1}^n \left\{ \sum_{r < c} y_{ir} \mathbf{x}_{(i)}^\top \boldsymbol{\beta}_r - m_i b(\boldsymbol{\eta}_i) \right\}, \quad \eta_{ir} = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}_r.$$

Then:

- (a) the score has blocks $\partial \ell / \partial \boldsymbol{\beta}_r = \sum_i (y_{ir} - m_i \pi_{ir}) \mathbf{x}_{(i)} = \mathbf{X}^\top (\mathbf{y}_r - \boldsymbol{\mu}_r)$, where $\boldsymbol{\mu}_r$ is the vector with entries $m_i \pi_{ir}$;
- (b) the observed and expected information coincide and have blocks $\mathbf{X}^\top \mathbf{W}_{rs} \mathbf{X}$ with $\mathbf{W}_{rs} = \text{diag}\{m_i \pi_{ir} (\delta_{rs} - \pi_{is})\}_{i=1}^n$;
- (c) ℓ is concave, and strictly concave when $\text{rank}(\mathbf{X}) = p$; in that case it has at most one maximizer, and the maximizer, if it exists, is the unique solution of the likelihood equations $\mathbf{X}^\top (\mathbf{y}_r - \boldsymbol{\mu}_r) = \mathbf{0}$ for $r = 1, \dots, c-1$.

Proof.

- (a) Differentiate: $\partial b(\boldsymbol{\eta}_i) / \partial \eta_{ir} = \pi_{ir}$ by Lemma 36.1.1, and $\partial \eta_{ir} / \partial \boldsymbol{\beta}_r = \mathbf{x}_{(i)}$.
- (b) Differentiating again, and using $\partial^2 b / \partial \eta_{ir} \partial \eta_{is} = \pi_{ir} (\delta_{rs} - \pi_{is})$,

$$-\frac{\partial^2 \ell}{\partial \boldsymbol{\beta}_r \partial \boldsymbol{\beta}_s^\top} = \sum_i m_i \pi_{ir} (\delta_{rs} - \pi_{is}) \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top.$$

The data enter only through the score, so the second derivative is nonrandom and observed information equals expected information. This is the canonical-link simplification of Theorem 34.3.1.

- (c) Take $\mathbf{a} = (\mathbf{a}_1^\top, \dots, \mathbf{a}_{c-1}^\top)^\top$ and put $t_{ir} = \mathbf{x}_{(i)}^\top \mathbf{a}_r$ for $r < c$ and $t_{ic} = 0$. By (b),

$$\mathbf{a}^\top (-\nabla^2 \ell) \mathbf{a} = \sum_i m_i \left\{ \sum_{r=1}^c \pi_{ir} t_{ir}^2 - \left(\sum_{r=1}^c \pi_{ir} t_{ir} \right)^2 \right\},$$

the inner brace being the variance of the random variable that takes the value t_{ir} with probability π_{ir} . A variance is nonnegative, so ℓ is concave. It vanishes only if $t_{i1} = \dots = t_{ic}$, and since $t_{ic} = 0$ this forces $\mathbf{x}_{(i)}^\top \mathbf{a}_r = 0$ for every i and every $r < c$, that is $\mathbf{X} \mathbf{a}_r = \mathbf{0}$. When $\text{rank}(\mathbf{X}) = p$ this gives $\mathbf{a} = \mathbf{0}$, so the Hessian is negative definite everywhere and ℓ is strictly concave. A strictly concave function has at most one stationary point, and a stationary point is its maximum. ■

Idea. The likelihood equations say $\mathbf{X}^\top \mathbf{y}_r = \mathbf{X}^\top \hat{\boldsymbol{\mu}}_r$ for each non-baseline category: *fitted counts reproduce observed counts in every direction spanned by a regressor*, so with an intercept each category's fitted total is its observed total. This is the canonical-link moment matching of Theorem 34.3.1, one copy per logit.

Because the information is the expected information, Newton–Raphson and Fisher scoring are the same algorithm, and by Theorem 34.4.1 each step is a weighted least squares fit, with block weights $\mathbf{W}_i = m_i(\text{diag}(\tilde{\boldsymbol{\pi}}_i) - \tilde{\boldsymbol{\pi}}_i \tilde{\boldsymbol{\pi}}_i^\top)$ and working response $\boldsymbol{\eta}_i + \mathbf{W}_i^{-1}(\mathbf{y}_i - m_i \boldsymbol{\pi}_i)_{1:c-1}$. The only change from Chapter 34 is that the weights are $(c-1) \times (c-1)$ blocks rather than scalars, so each step is a generalized rather than a diagonally weighted least squares problem (Theorem 21.3.1).

Warning (Existence). Concavity guarantees uniqueness, not existence. As for binary data, the maximum is attained unless the categories can be separated. Take $m_i = 1$ and $\text{rank}(\mathbf{X}) = p$ as in Theorem 36.1.2(c), with observation i in category y_i . If there are vectors $\mathbf{a}_1, \dots, \mathbf{a}_c$, not all equal, with $\mathbf{x}_{(i)}^\top (\mathbf{a}_{y_i} - \mathbf{a}_r) \geq 0$ for every i and r , then differentiating ℓ along that direction gives $\sum_i \sum_r \pi_{ir} \mathbf{x}_{(i)}^\top (\mathbf{a}_{y_i} - \mathbf{a}_r) \geq 0$ at *every* point of the ray, the fitted probabilities being positive throughout. By Theorem 36.1.2(c) the restriction of ℓ to the ray is strictly concave, so its derivative is strictly decreasing; were it zero somewhere it would be negative beyond, contradicting the display. It is therefore positive throughout, and the supremum is approached only at infinity. (Without the rank condition the derivative can vanish along the whole ray, which forces every $\mathbf{x}_{(i)}$ orthogonal to every difference $\mathbf{a}_s - \mathbf{a}_t$; the maximum is then attained, but not unique.) The binary case, with its proof and its remedies, is Theorem 35.3.1; the multinomial statement is Albert and Anderson (1984). With many categories and small cells separation is common, and the remedies are the same: a penalty as in Definition 27.2.1, or the equivalent proper prior of Proposition 30.1.4 — see Chapter 27 and Chapter 30.

36.1.4 Reading the coefficients

Two facts about Eq. 36.1.4 are easy to state and easy to misuse.

Proposition 36.1.3 — Changing the baseline

Let $\hat{\boldsymbol{\beta}}_1, \dots, \hat{\boldsymbol{\beta}}_{c-1}$ maximize the likelihood of Definition 36.1.1, and set $\hat{\boldsymbol{\beta}}_c = \mathbf{0}$. Fix any $s \in \{1, \dots, c\}$. Then:

- (a) for every pair of categories r, r' , $\log(\pi_{ir}/\pi_{ir'}) = \mathbf{x}_{(i)}^\top (\boldsymbol{\beta}_r - \boldsymbol{\beta}_{r'})$; in particular a model with baseline s is the same model, with coefficients $\boldsymbol{\beta}_r^{(s)} = \boldsymbol{\beta}_r - \boldsymbol{\beta}_s$;
- (b) refitting the model with baseline s gives $\hat{\boldsymbol{\beta}}_r^{(s)} = \hat{\boldsymbol{\beta}}_r - \hat{\boldsymbol{\beta}}_s$, the same maximized log-likelihood and the same fitted probabilities;
- (c) the estimated covariance matrix of $\hat{\boldsymbol{\beta}}^{(s)}$ obtained from the refitted model equals $\mathbf{L} \hat{\text{Cov}}(\hat{\boldsymbol{\beta}}) \mathbf{L}^\top$, where $\mathbf{L}$ is the linear map of (a); so standard errors of pairwise contrasts do not depend on which baseline was used.

Proof.

- (a) Subtract two instances of Eq. 36.1.4; the baseline cancels because $\log(\pi_{ir}/\pi_{ir'}) =$

$$\log(\pi_{ir}/\pi_{ic}) - \log(\pi_{ir'}/\pi_{ic}).$$

- (b) The map $\mathbf{L} : \boldsymbol{\beta} \mapsto \boldsymbol{\beta}^{(s)}$ defined by $\boldsymbol{\beta}_r^{(s)} = \boldsymbol{\beta}_r - \boldsymbol{\beta}_s$ for $r \neq s$ is linear, and it is invertible: from $\boldsymbol{\beta}^{(s)}$ one recovers $\boldsymbol{\beta}_r = \boldsymbol{\beta}_r^{(s)} + \boldsymbol{\beta}_s^{(s)}$ with $\boldsymbol{\beta}_s^{(s)} = \mathbf{0}$. By (a) the two parameterizations give exactly the same family of probability vectors $\boldsymbol{\pi}_i$, so the two likelihoods are the same function composed with $\mathbf{L}$. A reparameterization by a bijection carries maximizers to maximizers and leaves the maximum value and every fitted quantity unchanged.
- (c) Under a linear reparameterization $\boldsymbol{\psi} = \mathbf{L}\boldsymbol{\beta}$ the information transforms as $\mathbf{J}^{(s)} = \mathbf{L}^{-\top} \mathbf{J} \mathbf{L}^{-1}$, by the chain rule applied to the score. Hence $(\mathbf{J}^{(s)})^{-1} = \mathbf{L} \mathbf{J}^{-1} \mathbf{L}^\top$, which is exactly the delta-method covariance of $\mathbf{L}\hat{\boldsymbol{\beta}}$ — and the delta method is exact here because $\mathbf{L}$ is linear.

■

So the coefficients are log odds ratios: $\hat{\beta}_{rj}$ is the estimated change in $\log(\pi_r/\pi_c)$ per unit of regressor j , and differences of coefficients are log odds ratios between two non-baseline categories. The baseline is a choice of *display*, not of model. What is not invariant is the reading of a single coefficient, and here is the trap.

Proposition 36.1.4 — A coefficient is not a derivative of a probability

In Definition 36.1.1, with $\boldsymbol{\beta}_c = \mathbf{0}$ and x_j a continuous regressor,

$$\frac{\partial \pi_{ir}}{\partial x_{ij}} = \pi_{ir} \left(\beta_{rj} - \sum_{s=1}^c \pi_{is} \beta_{sj} \right). \quad (36.1.5)$$

So the c derivatives sum to zero, the derivative for category r is positive exactly when β_{rj} exceeds the probability-weighted average $\bar{\beta}_j(\mathbf{x}_{(i)}) = \sum_s \pi_{is} \beta_{sj}$, and the baseline, whose coefficient is zero by construction, has a strictly positive derivative whenever $\bar{\beta}_j(\mathbf{x}_{(i)}) < 0$.

Proof. Write $\pi_{ir} = e^{\eta_{ir}} / \sum_s e^{\eta_{is}}$ with $\partial \eta_{is} / \partial x_{ij} = \beta_{sj}$. The quotient rule gives

$$\frac{\partial \pi_{ir}}{\partial x_{ij}} = \pi_{ir} \beta_{rj} - \pi_{ir} \frac{\sum_s e^{\eta_{is}} \beta_{sj}}{\sum_s e^{\eta_{is}}} = \pi_{ir} \left(\beta_{rj} - \sum_s \pi_{is} \beta_{sj} \right).$$

Summing over r gives $\bar{\beta}_j - \bar{\beta}_j = 0$. The remaining statements read off the sign of the bracket, with $\beta_{cj} = 0$.

■

Since $\bar{\beta}_j(\mathbf{x})$ moves with $\mathbf{x}$, a fitted probability can rise over one range of the regressor and fall over another, whatever the sign of its coefficient. Read coefficients as odds ratios; read effects on probabilities off the fitted curves.

36.1.5 Travel mode choice

Example. (Choosing how to travel)

Two hundred and ten travellers on the Sydney–Melbourne corridor each chose one of four modes: air (58), train (63), bus (30) or car (59). The regressors are the household income in units of \$10,000 and the size of the travelling party. Take car as the baseline. Fitting [Definition 36.1.1](#) by Newton–Raphson from $\beta = \mathbf{0}$ takes 7 steps to full double precision, and gives the following coefficients, with standard errors from the inverse information in parentheses.

logit	intercept	income (per \$10,000)	party size
air vs car	0.9435 (0.5498)	0.0354 (0.1030)	−0.6006 (0.1992)
train vs car	2.4938 (0.5357)	−0.5731 (0.1184)	−0.3098 (0.1956)
bus vs car	1.9780 (0.6717)	−0.3033 (0.1322)	−0.9404 (0.3245)

Income does not distinguish air from car ($z = 0.34$) but moves people strongly away from the ground alternatives: each extra \$10,000 of household income multiplies the odds of train rather than car by $e^{-0.5731} = 0.5638$. Read through [Proposition 36.1.3](#), the same fit raises the odds of air over train by $e^{0.6085}$ per \$10,000. Party size pushes travellers towards the car, most sharply away from the bus. The likelihood ratio statistic against the intercept-only model is $2(-253.3408 + 283.7588) = 60.836$ on six degrees of freedom.

```
import numpy as np
import statsmodels.api as sm

MODES = ["air", "train", "bus", "car"]          # car is category 4, the baseline

raw = sm.datasets.modechoice.load_pandas().data
chosen = raw[raw["choice"] == 1].sort_values("individual").reset_index(drop=True)
y = chosen["mode"].astype(int).to_numpy()      # 1 air, 2 train, 3 bus, 4 car
n, c = len(y), 4
X = np.column_stack([np.ones(n), chosen["hinc"] / 10.0, chosen["psize"]])
Y = np.zeros((n, c))
Y[np.arange(n), y - 1] = 1.0                  # indicator matrix, baseline last

print("chosen mode counts:", Y.sum(axis=0))
```

[Run this code online](#)

The fit itself is [Theorem 36.1.2](#) written out: build the probabilities, form the score and the block information, solve, repeat.

```
def probs(X, B):
    """Fitted probabilities of the baseline-category logit model; B is p x (c-1)."""
    eta = np.column_stack([X @ B, np.zeros(len(X))]) # baseline has eta = 0
    eta = eta - eta.max(axis=1, keepdims=True)       # stabilize the
    exponentials
    E = np.exp(eta)
    return E / E.sum(axis=1, keepdims=True)
```

```

def loglik(X, Y, B):
    return float(np.sum(Y * np.log(probs(X, B))))

def mnlogit(X, Y, tol=1e-10, maxit=50):
    """Newton-Raphson (= Fisher scoring, the link is canonical) for baseline logits.
    """
    n, p = X.shape
    c = Y.shape[1]
    B = np.zeros((p, c - 1))
    for it in range(1, maxit + 1):
        P = probs(X, B)
        U = (X.T @ (Y[:, :c - 1] - P[:, :c - 1])).T.ravel()      # score, stacked
        by category
        J = np.zeros(((c - 1) * p, (c - 1) * p))                  # expected =
        observed information
        for r in range(c - 1):
            for s in range(c - 1):
                w = P[:, r] * ((r == s) - P[:, s])
                J[r * p:(r + 1) * p, s * p:(s + 1) * p] = X.T @ (w[:, None] * X)
        step = np.linalg.solve(J, U)
        B = B + step.reshape(c - 1, p).T
        if np.max(np.abs(step)) < tol:
            break
    return B, J, it

B, J, iters = mnlogit(X, Y)
se = np.sqrt(np.diag(np.linalg.inv(J))).reshape(c - 1, X.shape[1]).T

for r in range(c - 1):
    print(f"{MODES[r]:>6s} vs car: " + " ".join(f"{b:7.4f} ({s:.4f})" for b, s in
        zip(B[:, r], se[:, r])))
print(f"log-likelihood {loglik(X, Y, B):.4f} after {iters} Newton steps")

```

[Run this code online](#)

The script behind this section checks the output against the `MNLogit` routine of `statsmodels`, coefficient by coefficient. It also carries out the substitution of [Proposition 36.1.3](#), refitting with train as the baseline and confirming that subtracting the train column from the fitted coefficient matrix reproduces the refit and leaves every fitted probability unchanged.

```

# changing the baseline: subtract the train column (category 2) from every column
Btrain = B - B[:, [1]]
Btrain = np.delete(np.column_stack([Btrain, -B[:, 1]]), 1, axis=1)      # drop train
, add car
Ytrain = Y[:, [0, 2, 3, 1]]      # put train
last
Bcheck, _, _ = mnlogit(X, Ytrain)

print("air vs train, refitted :", np.round(Bcheck[:, 0], 6))
print("air vs train, subtracted:", np.round(Btrain[:, 0], 6))
print("fitted probabilities identical:", np.allclose(probs(X, Bcheck)[:, [0, 3, 1,
2]], probs(X, B)))

```

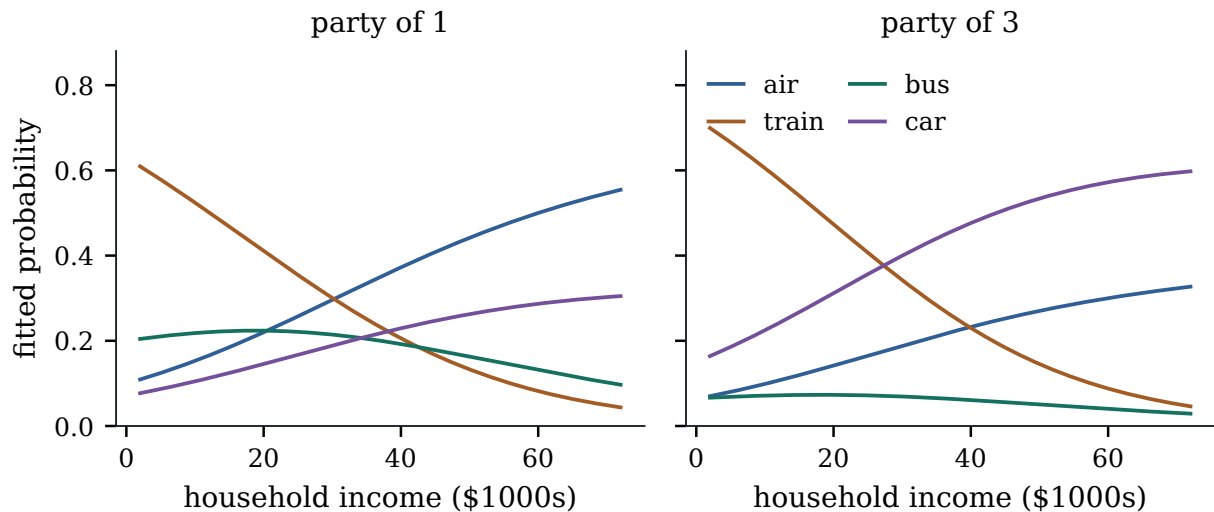


Figure 36.1.1: Fitted mode probabilities from the baseline-category logit model, against household income, for a traveller alone (left) and in a party of three (right). Coefficients are log odds ratios against the car; the curves show what those coefficients do to the probabilities themselves.

[Run this code online](#)

Figure 36.1.1 plots the fitted probabilities against income, and the bus curve in the left panel is Proposition 36.1.4 made visible: the coefficient of income in the bus logit is -0.3033 , yet the fitted probability of the bus *rises* over the first twenty thousand dollars before falling. At an income of \$10,000 with a party of one the fitted probabilities are (0.1528, 0.5243, 0.2182, 0.1047) for air, train, bus and car, with derivatives per \$10,000 of (0.0606, -0.1111 , 0.0126, 0.0378); the car, whose coefficient is fixed at zero, has the second largest of the four, behind air alone.

```
# the derivative of a fitted probability is not the coefficient
Bfull = np.column_stack([B, np.zeros(X.shape[1])]) # append the baseline's
# column of zeros
x0 = np.array([1.0, 1.0, 1.0]) # income $10,000, party
# of one
p0 = probs(x0[None, :], B)[0]
dp0 = p0 * (Bfull[1] - p0 @ Bfull[1]) # d pi_r / d (income in
# $10,000s)

print("fitted probabilities", np.round(p0, 4))
print("derivatives", np.round(dp0, 4))
print("income coefficients", np.round(Bfull[1], 4))
```

[Run this code online](#)

Remark (Fitting one logit at a time). It is tempting to estimate β_r by discarding every observation outside $\{r, c\}$ and running an ordinary binary logistic regression. By Eq. 36.1.4,

$$P(Y_i = r \mid Y_i \in \{r, c\}, \mathbf{x}_{(i)}) = \frac{\pi_{ir}}{\pi_{ir} + \pi_{ic}} = \frac{e^{\mathbf{x}_{(i)}^\top \beta_r}}{1 + e^{\mathbf{x}_{(i)}^\top \beta_r}},$$

so the conditional model is exactly the binary logistic model of Definition 35.1.1 with the same β_r , and the individualized fit is consistent. It is not efficient, and the $c - 1$ fits are correlated in a way the separate outputs do not report. Begg and Gray (1984) quantified the loss, small when the baseline is a large category and severe otherwise.

36.1.6 Inference

The argument of Theorem 34.5.1 carries over verbatim in block notation, although the theorem there is stated for a scalar response: by Theorem 36.1.2 the score is again a sum of independent mean-zero vectors, the information is again nonrandom, and the global concavity its proof leans on is Theorem 36.1.2(c). Under the analogues of its conditions — categories and regressors fixed, sample size growing, the leverages of the stacked design tending to zero, the true parameter interior — the maximum likelihood estimate is consistent and asymptotically normal with covariance the inverse information; nested models are compared by the analysis of deviance of Theorem 34.5.5.

Two cautions are specific to categorical responses. The deviance is *not* a goodness-of-fit statistic for ungrouped data: when every $m_i = 1$ the saturated model has as many parameters as observations and the χ^2 approximation fails, as Theorem 34.5.5 says in general and Proposition 35.4.1 for binary data. And a regressor is not one coefficient here but $c - 1$, so “does income matter?” has $c - 1$ degrees of freedom; testing the coefficients separately and reporting the smallest p -value is the multiplicity error of Chapter 13 in a new costume.

36.1.7 Exercises

36.1.7.1 A. Check your understanding

Exercise A1.

A response has $c = 5$ categories and the model matrix has $p = 4$ columns including the intercept. How many free coefficients has the baseline-category logit model? How many has the saturated model for n ungrouped observations? How many degrees of freedom does the test that one regressor may be dropped have?

Solution. The model has $(c - 1)p = 16$ free coefficients. The saturated model for ungrouped data gives each observation its own probability vector, $n(c - 1)$ free parameters. Dropping one regressor removes one coefficient from each of the $c - 1$ logits, so the test has $c - 1 = 4$ degrees of freedom.

36.1.7.2 B. Practice

Exercise B1.

From the fitted coefficients of Example (Choosing how to travel), compute the estimated log odds ratio of bus against train per \$10,000 of income, and the estimated odds ratio of air against bus per extra person in the party. State which of these depend on the choice of baseline.

Solution. By Proposition 36.1.3 the bus-against-train income coefficient is $-0.3033 - (-0.5731) = 0.2698$, so the odds of bus rather than train rise by a factor $e^{0.2698}$ per \$10,000. For party size, air against bus is $-0.6006 - (-0.9404) = 0.3398$, an odds ratio of $e^{0.3398}$. Neither depends on the baseline: by Proposition 36.1.3(a) a difference of coefficients is the pairwise log odds ratio $\log(\pi_{ir}/\pi_{ir'})$ per unit of the regressor, which the model specifies without reference to any baseline.

Exercise B2.

Show directly that the $c \times c$ matrix $\mathbf{\Sigma}(\boldsymbol{\pi}) = \text{diag}(\boldsymbol{\pi}) - \boldsymbol{\pi}\boldsymbol{\pi}^\top$ is nonnegative definite with $\mathbf{\Sigma}\mathbf{1} = \mathbf{0}$, and that its restriction to the first $c - 1$ coordinates is positive definite when every $\pi_r > 0$.

Solution. For any $\mathbf{t}$, $\mathbf{t}^\top \mathbf{\Sigma} \mathbf{t} = \sum_r \pi_r t_r^2 - (\sum_r \pi_r t_r)^2$ is the variance of a random variable taking the value t_r with probability π_r , hence nonnegative, and it vanishes iff all t_r with $\pi_r > 0$ are equal. Taking $\mathbf{t} = \mathbf{1}$ gives $\mathbf{\Sigma}\mathbf{1} = \mathbf{0}$. If $\mathbf{t}$ has $t_c = 0$ and $\mathbf{t}^\top \mathbf{\Sigma} \mathbf{t} = 0$, then all t_r are equal to $t_c = 0$; so the leading $(c - 1) \times (c - 1)$ block is positive definite.

36.1.7.3 C. Going deeper**Exercise C1.**

The **adjacent-category logit model** for an ordered response specifies $\log(\pi_{i,r+1}/\pi_{ir}) = \theta_r + \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ for $r = 1, \dots, c - 1$, with $\mathbf{x}_{(i)}$ not containing an intercept. Show that it is the baseline-category model [Definition 36.1.1](#) with slope vectors $\boldsymbol{\beta}_r = (c - r) \boldsymbol{\beta}^*$ for a single vector $\boldsymbol{\beta}^*$, plus a free intercept for each r : a submodel obtained by $(c - 2)p$ linear restrictions, fitted with the same software after recoding the model matrix.

Solution. Summing the adjacent logits from r to $c - 1$ gives $\log(\pi_{ic}/\pi_{ir}) = \sum_{s=r}^{c-1} \theta_s + (c - r) \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$, so $\log(\pi_{ir}/\pi_{ic}) = -\sum_{s \geq r} \theta_s - (c - r) \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$. This is [Eq. 36.1.4](#) with a free intercept for each r and slope vector proportional to $\boldsymbol{\beta}^* = -\boldsymbol{\beta}$ with the known factor $c - r$. Fitting is a baseline-category fit on the model matrix whose slope block for logit r is $(c - r)\mathbf{X}$.

Exercise C2.

Show that in [Definition 36.1.1](#) the ratio $\pi_{ir}/\pi_{ir'}$ does not involve any third category — the *independence of irrelevant alternatives*. Construct a three-category example in which this is substantively absurd, and say what the assumption forces about the fitted probabilities when a new category is added that is nearly a copy of an existing one.